

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 30

HALF ADDER USING NAND GATES

AIM: To design and implement a Half Adder using NAND gates circuit using Logisim simulator.

ALGORITHM

1. Open the Logisim simulator.
2. Place two Input Pins and name them A and B.
3. Place five NAND gates.
4. Connect A and B to the first NAND gate.
5. Use NAND gates to obtain the XOR operation for the Sum.
6. Use another NAND gate as an inverter to obtain the Carry.
7. Connect the outputs to Sum (S) and Carry (C).
8. Test all possible input combinations.
9. Verify the outputs with the truth table

Truth Table:

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

RESULT:

logisim.app

File Edit Project Simulate Window Help

Wiring

Splitter

Pin

Probe

Tunnel

Pull Resistor

Clock

Constant

Power

Ground

Transistor

Transmission Gate

Bit Extender

Pin

Facing	East
Output?	No
Data Bits	1
Three-state?	Yes
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12

0

0

0

AND

OR

0

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